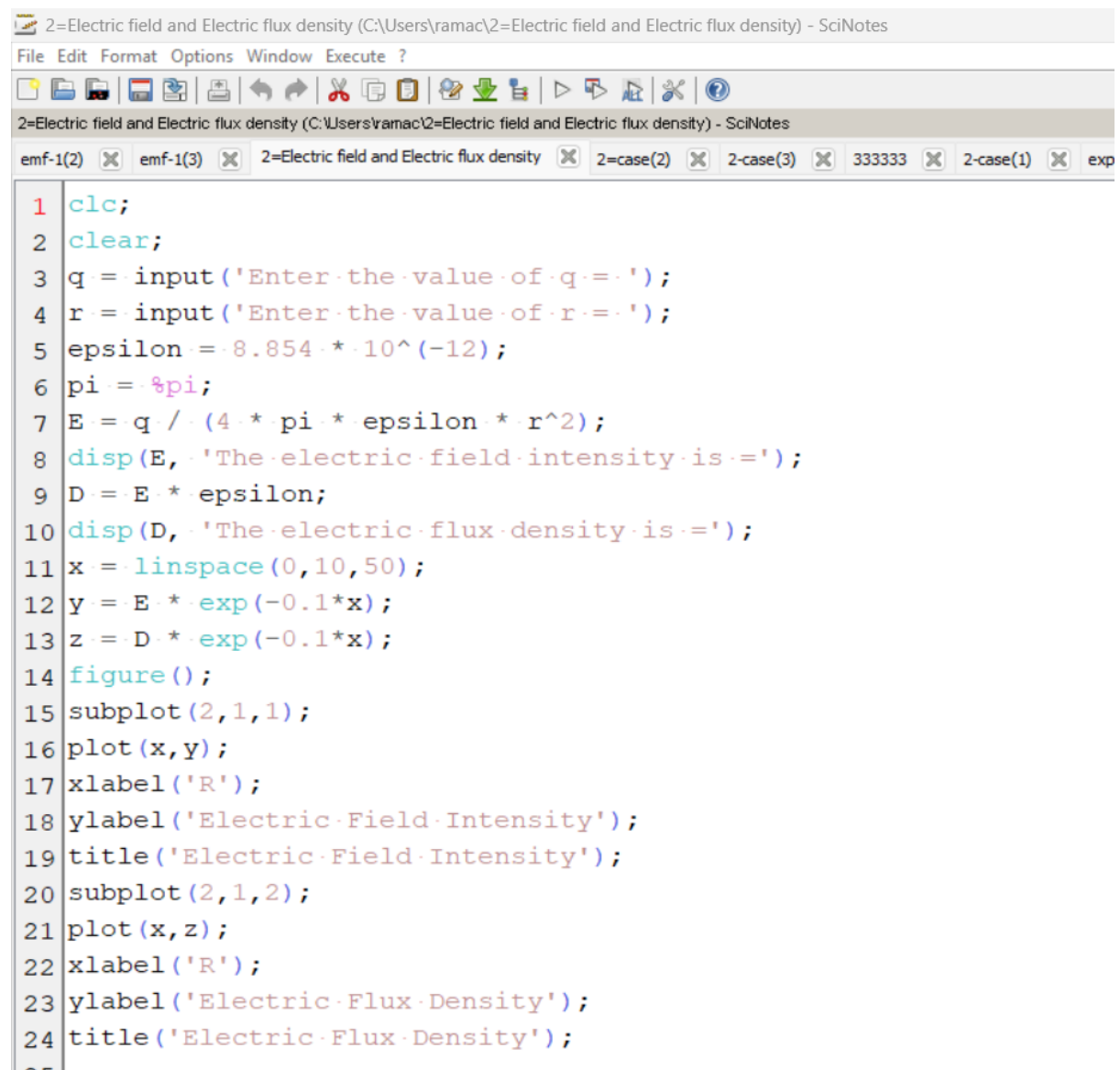


Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim: To write program for electric field and electric flux density in sphere using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



```
2=Electric field and Electric flux density (C:\Users\ramac\2=Electric field and Electric flux density) - SciNotes
File Edit Format Options Window Execute ?
2=Electric field and Electric flux density (C:\Users\ramac\2=Electric field and Electric flux density) - SciNotes
emf-1(2) emf-1(3) 2=Electric field and Electric flux density 2=case(2) 2=case(3) 333333 2=case(1) exp

1 clc;
2 clear;
3 q = input('Enter the value of q = ');
4 r = input('Enter the value of r = ');
5 epsilon = 8.854 * 10^(-12);
6 pi = %pi;
7 E = q ./ (4 * pi * epsilon * r^2);
8 disp(E, 'The electric field intensity is = ');
9 D = E * epsilon;
10 disp(D, 'The electric flux density is = ');
11 x = linspace(0,10,50);
12 y = E * exp(-0.1*x);
13 z = D * exp(-0.1*x);
14 figure();
15 subplot(2,1,1);
16 plot(x,y);
17 xlabel('R');
18 ylabel('Electric Field Intensity');
19 title('Electric Field Intensity');
20 subplot(2,1,2);
21 plot(x,z);
22 xlabel('R');
23 ylabel('Electric Flux Density');
24 title('Electric Flux Density');
```

```
Scilab 6.1.1 Console

File Edit Control Applications ?

[Icons]

Scilab 6.1.1 Console

Enter the value of q = 4*10^6

Enter the value of r = 3*10^-2

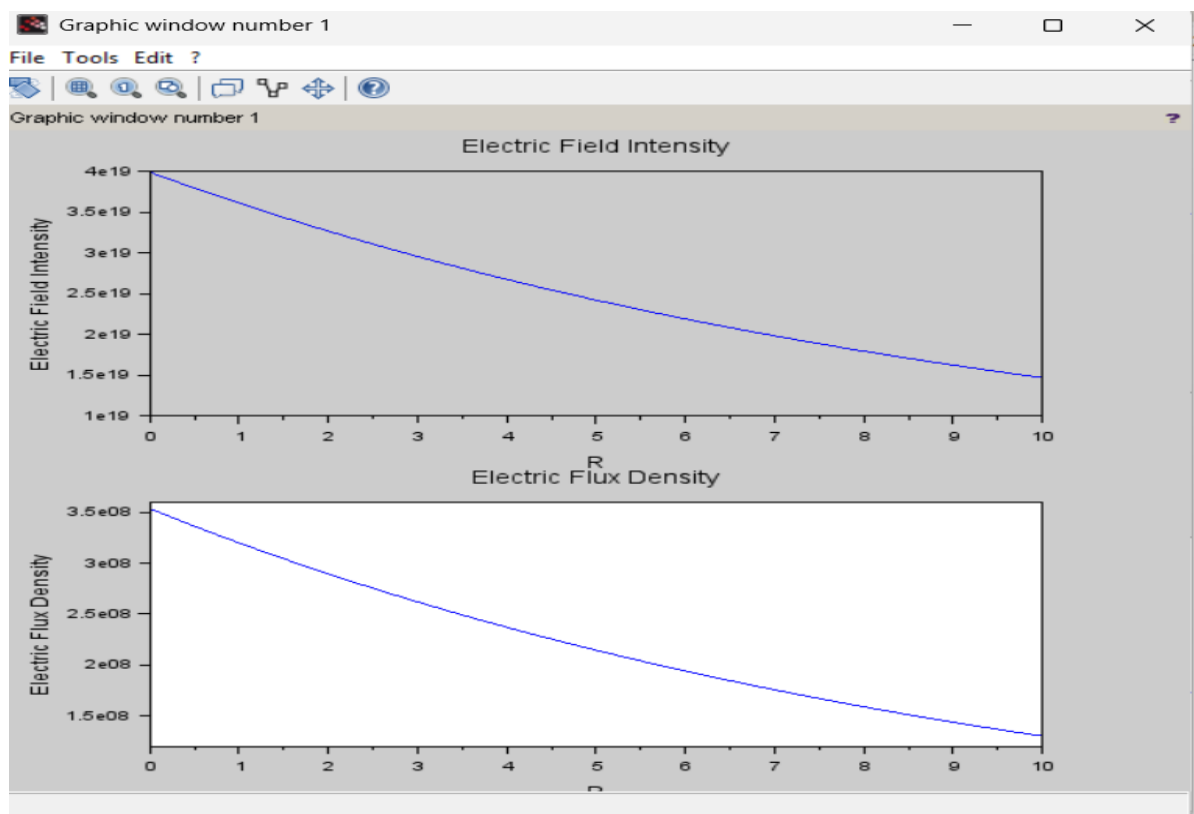
3.995D+19

"The electric field intensity is ="

3.537D+08

"The electric flux density is ="

-->
```



Theoretical output:

Handwritten calculations on a piece of paper:

1) Electric field intensity

$$E = \frac{Q}{4\pi\epsilon r^2}$$
$$= \frac{4 \times 10^{-6}}{4 \times \pi \times 8.85 \times 10^{-12} \times (1)^2}$$
$$\Rightarrow \frac{4 \times 10^{-6}}{1.112} \Rightarrow 3.59 \times 10^6$$

2) Electric flux density

$$D = E \epsilon$$
$$= (3.59 \times 10^6) \times (8.85 \times 10^{-12}) \text{ N/m}$$
$$\Rightarrow 3.177 \times 10^{-17} \text{ C/m}^2$$

There are red checkmarks and a red arrow pointing from the first calculation to the second.

Result :

The **electric field intensity (E)** and **electric flux density (D)** of a uniformly charged sphere were successfully determined and plotted for different values of radius. The plots showed that the electric field and flux density vary with the distance from the centre of the sphere and change according to the radius of the charged sphere.